

Sample input/output

This was the input books inserted in the trees

Random Books (30 book):

17,Deep Learning,Ian Goodfellow
3,Introduction to Algorithms,Thomas Cormen
25,Hands-On Machine Learning,Aurelien Geron
1,Clean Code,Robert Martin
12,Refactoring,Martin Fowler
29,The Art of Computer Programming,Donald Knuth
8,Computer Networks,Andrew Tanenbaum
21,Computer Organization and Design,David Patterson
5,Effective C++,Scott Meyers
14,The Mythical Man-Month,Frederick Brooks
2,The Pragmatic Programmer,Andrew Hunt
30,Think Python,Allen Downey
9,Operating System Concepts,Abraham Silberschatz
18,Compilers,Alfred Aho
6,C++ Primer,Stanley Lippman
27,Software Engineering,Ian Sommerville
11,Modern Operating Systems,Andrew Tanenbaum
4,Design Patterns,Erich Gamma
23,Introduction to Machine Learning,Ethem Alpaydin
15,Code Complete,Steve McConnell
28,Data Structures and Algorithms in C++,Michael Goodrich
7,Artificial Intelligence,A Stuart Russell
19,Cracking the Coding Interview,Gayle Laakmann
10,Database System Concepts,Henry Korth
24,Python Crash Course,Eric Matthes
13,Head First Java,Kathy Sierra
26,Grokking Algorithms,Aditya Bhargava
16,Structure and Interpretation of Computer Programs,Harold Abelson
22,The C Programming Language,Dennis Ritchie
20,Programming Pearls,Jon Bentley

Sorted Books (30 book):

5,The Silent Ocean,Michael Reed
8,Shadows of Tomorrow,Emma Carter
11,Data Structures Simplified,Daniel Moore
14,The Lost Kingdom,Sophia Turner
17,Algorithms in Practice,James Walker
20,Mystery at Blackwood,Liam Harris
23,Cloud Computing Basics,Olivia Scott
26,Journey Beyond Stars,Noah Bennett
29,The Last Empire,Isabella Young
32,Machine Learning Essentials,Ethan Hall
35,Secrets of the Nile,Mia Adams
38,Modern C++ Guide,Alexander King
41,Whispers in the Wind,Charlotte Green
44,Database Design Concepts,Benjamin Clark
47,The Hidden Truth,Amelia Baker
50,Artificial Intelligence Today,William Lewis
53,The Forgotten Path,Evelyn Wright
56,Operating Systems Explained,Henry Lopez
59,The Crystal Island,Abigail Hill
62,Networking Fundamentals,Lucas Rivera
65,Legends of Avalon,Harper Mitchell
68,Software Engineering Principles,Elijah Perez
71,The Dark Horizon,Scarlett Roberts
74,Cybersecurity Essentials,Jack Turner
77,The Ancient Code,Grace Phillips
80,Web Development Mastery,Sebastian Campbell
83,Echoes of Eternity,Chloe Parker
86,Python for Beginners,Matthew Evans
89,The Midnight Secret,Lily Edwards
92,Advanced Database Systems,David Collins

Output:

Random IDs

BST height: 7

Number of steps finding 8: 4

AVL height: 6

Number of steps finding 8: 2

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Sorted IDs

BST height: 30

Number of steps finding 65: 21

AVL height: 5

Number of steps finding 65: 5

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Random IDs
BST height: 7
Number of steps finding 8: 4
-----
AVL height: 6
Number of steps finding 8: 2
=====
Sorted IDs
BST height: 30
Number of steps finding 65: 21
-----
AVL height: 5
Number of steps finding 65: 5
```